Higher National Diploma in Engineering
Fourth Year, First Semester Examination - 2023
EE3204 - Electrical Energy Management

Instructions for Candidate:

Answer any **Five (05)** questions
All questions carry equal marks

No. of Questions: 06
No. of Pages: 04
Time : 3 (hours)

1.

i.) Describe the construction of shell type transformer and compare the voltage regulation and efficiency of core type and shell type transformers

[3 marks]

ii.) Briefly explain the vector groups of three phase transformers and discuss its importance of considering the vector group of a transformer

[4 marks]

iii.) Draw the primary referred exact equivalent circuit for the transformer and clearly name the circuit parameters

[3 marks]

iv.) Industrial process requires 2000V and 400V voltages and for that purpose there is old transformer installed. This transformer is 10.5 kVA, 2000/400 V single phase transformer and it have primary referred equivalent resistance of 6.1 Ω and equivalent reactance of 10.6 Ω. Transformer operated with full load and 0.8 lagging power factor. Core loss of the transformer is 250 W;

a.) Calculate the efficiency of this transformer

b.) What is the voltage regulation of this transformer

c.) It is proposed to replace this old transformer with a new transformer and it is 10.5 kVA, 2000/400 V, single phase transformer with primary referred resistance of 1.9 Ω and reactance of 5.1 Ω. Calculate the efficiency and voltage regulation of this new transformer if the core loss of the transformer is 220 W

[4+2+4 = 10 marks]

2.

i.) Define following quantities related to lighting;

a.) Luminous efficacy
b.) Luminous intensity
c.) Illumination

[6 marks]

ii.) One of the important pieces of information provided by light source manufacturers is, luminous **intensity distribution curve** of the light source. Describe what is the luminous intensity **distribution curve** and describe its importance in lighting design

[3 marks]

iii.) Explain the parameter, **color rendering index (CRI)** of a light source and compare the CRI values for incandescent bulb, fluorescent bulb and LED lamps

[3 marks]

iv.) Room is 16 m long and width of the room is 10 m. General illumination of 200 lux need to be maintained in the room, and height to the working plane is 2.5 m from the lighting fixtures, Consider the utilization factor of 0.8 and maintenance factor of 0.7 and find following;

a.) If fluorescent tube lights with 80W and efficacy of 55 lm /W are used, how many lights are required to provide the expected illumination (space to height ratio is 1.5)

b.) What is the spacing between light sources. Draw the arrangement

[5+3 = 8 marks]

3.

i.) Briefly describe the differences between wound rotor induction motor and squirrel cage induction motor and why wound rotor induction motors are more efficient

[4 marks]

ii.) Deep bar rotor induction motors have **high starting torque and high full load efficiency**. Explain how the deep bar rotor construction allows the motor to have above mentioned features (high starting torque and high full load efficiency)

[5 marks]

iii.) Draw the torque-slip characteristics for 3-phase induction motor

[3 marks]

iv.) A 4-pole three 400 V, 11 kW, 50 Hz delta-connected induction motor has the following impedances in ohms per phase referred to the stator circuit:

$R_1 = 0.2\,\Omega \quad R_2' = 1.72\,\Omega$

$X_1 = 4.6\,\Omega \quad X_2' = 7.1\,\Omega$

The total rotational losses are 700 W and core losses are 400 W. For a rotor slip of 8% at the rated voltage and rated frequency, find the following;

a.) Speed of the motor
b.) Stator current
c.) Power factor
d.) Efficiency of the motor

[8 marks]

4.

i.) Briefly describe the following types of tariffs;

a.) Block rate tariff
b.) Time of day tariff

[4 marks]

ii.) A generating station has a maximum demand of 70 MW and yearly load factor of 65%. Fixed cost in generation is 40 million LKR, transmission fixed cost is 20 million LKR and fixed distribution cost is 25 million LKR, per annuum. Respective diversity factors for transmission and distribution are 1.1 and 1.25. Efficiency of the transmission system is 96% and efficiency of distribution is 90%. Fuel cost for generation is 850 million LKR. Find the yearly average cost per kW demand and per kWh;

a.) At substation

b.) At consumer premises

[4+3 = 7 marks]

iii.) Briefly describe the power factor correction using synchronous condenser method and what the advantages of this method compared to using capacitor banks to power factor improvement

[4 marks]

iv.) Load of an industrial premises is 675 kW, 0.775 power factor lagging. This premises works for 2920 hours for annuum. Tariff is maximum demand tariff and rates are 1050 LKR/kVA maximum demand and 25 LKR/kWh for energy charges. If the power factor is improved to 0.95 PF lagging by means of a loss free capacitors costing 11,000 LKR/kvar. Calculate the annual cost saving due to power factor improvement

[5 marks]

5.

i.) Briefly describe the importance of energy management

[4 marks]

ii.) Explain the following load management techniques and give two activities related to each of these methods;

a.) Supply side management (SSM)

b.) Demand side management (DSM)

[6 marks]

iii.) Define the term **specific energy consumption** and explain how this value can be determined for some company or industry

[4 marks]

iv.) State **four** efficient lighting design principles and write short note on each of those methods

[6 marks]

6.

i.) State **five** responsibilities of an energy manager

[5 marks]

ii.) Explain the term **energy accounting** and describe the activities related to energy accounting

[4 marks]

iii.) What are the main objectives of conducting an energy audit

[2 marks]

iv.) Briefly explain the requirement of a detailed energy audit and state the steps involved with a detailed energy audit

[5 marks]

.) Briefly explain the qualities required to have for good energy auditor

[4 marks]